Decent Coaching Classes

VIII SB

Mind Your Math VIII

Time: 1 Hour **Marks: 30**

Q.1 (A) Four alternatives are given for each sub-question. Choose the correct one. **(3)**

(1) The point of concurrence of the three altitudes of a triangle is called the

(A) centroid (B) incentre (C) orthocentre (D) circumcentre

(2) If x varies inversely as y, then which of the following is constant?

(A) x / y (B) $x \times y$ (C) $x + y$ (D) $x - y$

(3) The factors of $x^2 - 5x + 6$ are

(A) $(x + 2)(x + 3)$ (B) $(x - 1)(x - 6)$ (C) $(x + 6)(x - 1)$ (D) $(x - 2)(x - 3)$

Q.1 (B) Solve the following sub-questions. **(2)**

(1) G is the centroid of Δ ABC and seg AD is a median. If l(AD) = 12 cm, find l(AG).

(2) y varies directly as x. When x = 4, y = 20. Find the constant of variation.

Q.2 Solve any four of the following sub-questions. **(8)**

(1) Factorise: $x^2 + 9x + 20$

(2) Factorise: $x^3 - 27$

(3) Write the following statement using the symbol of variation and as an equation: "The area (A) of a circle is directly proportional to the square of its radius (r)."

(4) Draw an obtuse-angled Δ LMN. Draw all its medians and show the centroid G.

(5) a varies inversely as b. When a = 10, b = 6. Find the value of a when b = 4.

(6) Simplify: $(x^2 - 9) / (x + 3)$

Q.3 Solve any three of the following sub-questions. **(9)**

(1) Factorise: $2x^2 - 7x - 15$

(2) Factorise: $54p^3 + 250q^3$

(3) A bus travelling at a speed of 45 km/hr takes 4 hours to complete a journey. How much time will it take to cover the same distance at a speed of 60 km/hr?

(4) The cost of sugar is directly proportional to its weight. If the cost of 5 kg of sugar is ₹ 225, find the cost of 12 kg of sugar.

(5) G is the centroid of Δ ABC. Seg BQ and seg CR are medians. If l(BG) = 10 cm and l(GR) = 3.5 cm, find l(GQ) and l(CR).

Q.4 Solve any two of the following sub-questions. **(8)**

(1) Simplify: $(x^2 - 9)/(x^2 + 5x + 6) \times (x^2 + 2x)/(x^2 - 3x)$

(2) (a) Factorise: $6x^2 + 7x - 3$

(b) Factorise: $125a^3 - 8b^3$

(3) Draw Δ XYZ with l(XY) = 5 cm, l(YZ) = 6 cm and l(XZ) = 7 cm. Draw all its altitudes and show the orthocentre O.